

Modern Math (PNC) : —

* Combination → No. of ways to select (selection)

$$1) {}^nC_r = \frac{n!}{r! (n-r)!} \quad \left| \text{Ex} - {}^5C_2 = \frac{5!}{2! 3!} \right.$$

* Permutation → selection + Arrangement

$$2) {}^nP_r = {}^nC_r \times r! \quad \left| \text{Ex} :- {}^5P_2 = \frac{5!}{3!} \right.$$

$$\text{or } {}^nP_r = \frac{n!}{(n-r)!}$$

* In how many ways a person can travel from Ko to Delhi via Jaipur



B₁ → 10 ways

B₂ → 10 ways

B₃ → 10 ways

⋮
B₁₀ → 10 ways

Total = 80 ways

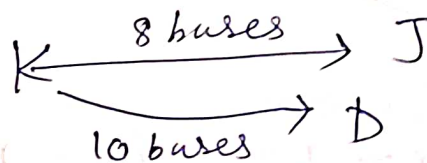
$$\text{Ans} \Rightarrow 8 \times 10 = 80 \text{ ways}$$

[and → ×]

* In how many ways a person can travel to —

1) Jaipur or Delhi

2) Jaipur and Delhi



$$\text{Answer} \rightarrow 1) 10 + 8 = 18 \quad (\text{OR} \rightarrow +)$$

$$\text{Answer} \rightarrow 2) 10 \times 8 = 80 \quad (\text{AND} \rightarrow \times)$$

* > Ordered no. solution / whole no. of solution

Formula — $(n+r-1)C_{r-1}$ [similar things → different persons]

Ex: — $a+b+c=10$

$$\begin{aligned} \rightarrow 10+3-1C_{3-1} &= 12C_2 = \frac{12!}{2!10!} \\ &= \frac{12 \times 11 \times 10!}{2 \times 10!} \\ &= 66 \checkmark \end{aligned}$$

* > find natural no solutions :-

$$\begin{aligned} a+b+c &= 10 \\ \rightarrow a+b+c &= 7 \end{aligned}$$

$$\rightarrow 7+3-1C_{3-1} = 9C_2 = \frac{9 \times 8}{2} = 36$$

* > they will give restriction . find ordered solution

1) $\left\{ \begin{array}{l} a+b+c=10 \text{ (given)} \\ 1 \leq a \leq 3 \text{ (given)} \end{array} \right\}$ Transform equation

$\rightarrow \boxed{a+b+c=9} ; \boxed{0 \leq a \leq 2}$

$$\rightarrow 9+3-1C_{3-1} = 11C_2 = 55$$

2) Remove cases :-

$\boxed{a \geq 3} ; \boxed{a+b+c=9} \rightarrow a+b+c=6$

$$\therefore 6+3-1C_{3-1} = 8C_2 = 28$$

$$\therefore \text{Answer} = 55 - 28 = 27 (\text{Ans})$$

*) Restriction on 3 . find ordered solution

Ex: — $A + B + C = 16$

$$\begin{aligned} 1 &\leq A \leq 6 \\ 1 &\leq B \leq 6 \\ 1 &\leq C \leq 6 \end{aligned}$$

Step 1) Transformation

$$\rightarrow \boxed{A + B + C = 13} \rightarrow {}^{13+3-1}C_{3-1} = {}^{15}C_2 = 105 \text{ Cases}$$

$$\begin{aligned} 0 &\leq A \leq 5 \\ 0 &\leq B \leq 5 \\ 0 &\leq C \leq 5 \end{aligned}$$

Step 2) Removal of cases —

1) $A \geq 6 \rightarrow A + B + C = 13 \rightarrow {}^9C_2 = 36 \text{ cases}$
 $A + B + C = 7$

2) $B \geq 6 \rightarrow A + B + C = 13 \rightarrow {}^9C_2 = 36 \text{ cases}$
 $A + B + C = 7$

3) $C \geq 6 \rightarrow A + B + C = 13 \rightarrow {}^9C_2 = 36 \text{ cases}$
 $A + B + C = 7$

4) $A \geq 6 \& B \geq 6 \rightarrow A + B + C = 13 \rightarrow {}^3C_2 = 3 \text{ cases}$
 $A + B + C = 1$

5) $A \geq 6 \& C \geq 6 \rightarrow A + B + C = 13 \rightarrow 3 \text{ cases}$
 $A + B + C = 1$

6) $B \geq 6 \& C \geq 6 \rightarrow A + B + C = 13 \rightarrow 3 \text{ cases}$
 $A + B + C = 1$

7) $A \geq 6, B \geq 6, C \geq 6 \rightarrow A + B + C = 13 \rightarrow 0 \text{ cases}$

Total = $A + B + C - (A \cap B + A \cap C + B \cap C) + (A \cap B \cap C)$

Removal = $36 + 36 + 36 - (3 + 3 + 3) + 0$
 $= 99$

∴ Ans = $105 - 99 = 6 \text{ (Ans)}$

* Dice concept $\rightarrow$ 3 dices $\rightarrow$ min sum = 3, max sum = 18

find no of ways in which a score of 15 can be made from the throw by three persons

$$A + B + C = 15 \quad \checkmark$$

$$1 \leq A \leq 6 \rightarrow A = 6 - x ; 0 \leq x \leq 5$$

$$1 \leq B \leq 6 \rightarrow B = 6 - y ; 0 \leq y \leq 5$$

$$1 \leq C \leq 6 \rightarrow C = 6 - z ; 0 \leq z \leq 5$$

Substituting:

$$(6 - x) + (6 - y) + (6 - z) = 15$$

$$\Rightarrow x + y + z = 3 \quad \checkmark$$

$$\text{Ans} \Rightarrow {}^{3+3-1}C_{3-1} = {}^5C_2 = \frac{5 \times 4}{2} = 10 \text{ (Ans)}$$

* Unordered solution (Example 1)

$$a + b + c = 10$$

$$\rightarrow \text{ordered solution} = {}^{12}C_2 = \frac{12 \times 11}{2} = 66$$

$$\therefore \boxed{6x + 3y + z = 66}$$

Wow, 2 same & 1 different cases —

$$\underline{C_2} \Rightarrow \text{where } a = b, 2a + c = 10 \therefore a = \{0, 1, 2, 3, 4, 5\}$$
$$\Rightarrow c = 10 - 2a \therefore y = 6$$

C3) All three same —

where $a = b = c$ — No cases

$$\therefore z = 0$$

$$\text{Thus, } 6x + (3 \times 6) + 0 = 66$$

$$\Rightarrow 6x = 66 - 18 \Rightarrow 6x = 48$$
$$\Rightarrow x = 8$$

Unordered solution

$$x + y + z$$

$$= 8 + 6 + 0 = 14 \text{ (Ans)}$$

Unordered Solution (Example-2)

In how many ways $\overset{\text{(similar)}}{15 \text{ identical Apples}}$ can be distributed among $\underset{\text{(different)}}{A, B, C}$ s.t. $\underset{\text{(all distinct)}}{C > B > A}$ & all will get at least 2 Apples

Ans $\rightarrow A + B + C = 15$

$A \geq 2$

$B \geq 2$

$C \geq 2$

Transform $\rightarrow$

$$A + B + C = 9$$

$$\therefore \text{Total solution} = {}^{9+3-1}C_{3-1} = 55$$

$$\therefore 6x + 3y + z = 55$$

Now, 2 same & 1 different

where $A = B \rightarrow 2A + C = 9 \left[\because A = 0 \text{ to } 4 \right]$

$$\Rightarrow C = 9 - 2A$$

$$y = 5 \text{ cases}$$

$$\begin{array}{r} - 1 \\ \hline 4 \text{ cases} \end{array}$$

Now, all same

$$A = B = C, A + B + C = 9$$

$$\Rightarrow 3A = 9$$

$$\Rightarrow A = 3$$

$$\therefore A, B, C = 3$$

$$z = 1 \text{ case}$$

Now, $6x + 3y + z = 55$

$$\Rightarrow 6x + (3 \times 4) + 1 = 55$$

$$\Rightarrow 6x = 42 \Rightarrow x = 7$$

If A, B, C distinct $\rightarrow \text{Ans} = 6x = 42$

If A, B, C not distinct $\rightarrow \text{Ans} = x = 7$

* Boys & Girls Concept

Example 1:— 6 Boys, 4 girls

a) All boys together ———

$\boxed{B_1 B_2 B_3 B_4 B_5 B_6} \quad G_1 \quad G_2 \quad G_3 \quad G_4$

$$\therefore \text{No. of ways} = \underline{5} \times \underline{6}$$

b) No girls are together ———

$\boxed{G_1 G_2 G_3 G_4} \quad B_1 B_2 B_3 B_4 B_5 B_6$

$$\therefore \text{No. of ways} = \underline{7} \times \underline{4}$$

c) No two girls are together/Consecutive ———

— B_1 — B_2 — B_3 — B_4 — B_5 — B_6 —

$$\text{No. of ways} = {}^7C_4 \times \underline{4} \times \underline{6}$$

d) No girls are together ——— (fix Boys): ———

— B_1 — B_2 — B_3 — B_4 — B_5 — B_6 — $\therefore \text{No. of ways} = {}^7C_4 \times \underline{4} \times \underline{6}$

OR — (Distribution method) $\rightarrow$ (fix girls): ———

$\textcircled{a} G_1 \textcircled{u} G_2 \textcircled{y} G_3 \textcircled{z} G_4 \textcircled{b}$
 $\geq 0 \quad \geq 1 \quad \geq 1 \quad \geq 1 \quad \geq 0$

$$a + u + y + z + b = \text{Total Boys} = 6$$

$$\rightarrow a + u + y + z + b = 3 \rightarrow {}^{3+5-1}C_4 \times \underline{4} \times \underline{6} \\ = {}^7C_4 \times \underline{4} \times \underline{6}$$

e) All Boys are together, all girls are together ———

$\boxed{B_1 B_2 B_3 B_4 B_5 B_6} \quad \boxed{G_1 G_2 G_3 G_4} = \underline{2} \times \underline{6} \times \underline{4}$

*) A B C D E (given) is a word.

Total ways of arrangement = $5! = 120$

a) No. of ways of arrangement when A comes before B ($A > B$) :-

A B ✓ 2 cases $\rightarrow$ 1 case

B A ✗ 1 $\rightarrow$ $1/2$

$120 \rightarrow 120 \times \frac{1}{2} = 60$ cases / ways

b) No of ways where A comes before B, who is in front of C ($A > B > C$)

A B C ✓ 6 cases $\rightarrow$ 1 case

A C B ✗ 1 $\rightarrow$ $1/6$ case

B A C ✗ $120 \rightarrow 120 \times \frac{1}{6} = 20$ ways

B C A ✗

C A B ✗

C B A ✗

c) No. of ways where A comes in between B & C.

A B C ✗ 6 cases $\rightarrow$ 2 case

A C B ✗ 1 case $\rightarrow$ $6/2 = 1/3$

C A B ✓ $120 \rightarrow 120 \times \frac{1}{3} = 40$ ways

C B A ✗

B A C ✓

B C A ✗

* Number concept

a) Make a 4 digit no (Repetition allowed)

$$\begin{array}{cccc} a & b & c & d \\ 9 & 10 & 10 & 10 \end{array} \quad \text{No of ways} = 9 \times 10^3 \text{ ways}$$

Make a 4 digit no. (Repetition not allowed)

$$\begin{array}{cccc} a & b & c & d \\ 9 & 9 & 8 & 7 \end{array} \quad \text{No of ways} = 9 \times 9 \times 8 \times 7 \text{ ways}$$

b) Make a 3 digit even no (Repetition allowed)

$$\begin{array}{ccc} a & b & c \\ \downarrow & \downarrow & 0/2/4/6/8 \\ 9 & 10 & 5 \end{array} \quad \text{No of ways} = 9 \times 10 \times 5 \text{ ways}$$

c) Make a 3 digit even no. (Repetition not allowed)

1) $\begin{array}{ccc} a & b & c \\ \downarrow & \downarrow & 0 \\ 9 & 8 & 2 \end{array}$ No. of ways = $9 \times 8 \times 1 = 72 \text{ ways}$
(fix zero) +

2) $\begin{array}{ccc} a & b & c \\ \downarrow & \downarrow & 2/4/6/8 \\ 9 & 8 & 4 \end{array}$ No. of ways = $9 \times 8 \times 4 = 288 \text{ ways}$

∴ Total No of ways = $72 + 288 = 360 \text{ ways}$
(fix 2, 4, 6, 8)

d) How many even digits b/w 100 & 700 with distinct digits

$$\begin{array}{cccc} a & b & c & d \\ 1/3/5/6 & & 0/2/4/6/8 & \end{array} \quad \text{Common case} \rightarrow 4, 6$$

c1) fix 4 $\div$ $\begin{array}{cccc} a & b & c & d \\ 5/6 & & & 4 \\ 2 & 8 & 7 & 1 \end{array}$ No of ways = $2 \times 8 \times 7 \times 1 = 112$
+

c2) fix 6 $\div$ $\begin{array}{cccc} a & b & c & d \\ 1/5 & & & 6 \\ 2 & 8 & 7 & 1 \end{array}$ No of ways = $2 \times 8 \times 7 \times 1 = 112$
+

c3) Remaining $\div$ $\begin{array}{cccc} a & b & c & d \\ 1/3/5/6 & & 0/2/8 & \\ 3 & 8 & 7 & 3 \end{array}$ No of ways = $3 \times 8 \times 7 \times 3 = 504$
Case Ans = 728 ways

e) Make a 4 digit odd no (Repetition allowed) —

a	b	c	d	
↓	↓	↓		1/3/5/7/9
⑨	⑩	⑩	⑤	No of ways = $9 \times 10 \times 10 \times 5$ ways

make a 4 digit odd no (Repetition not allowed) —

a	b	c	d	
↓	↓	↓		1/3/5/7/9
⑧	⑧	⑦	⑤	No of ways = $8 \times 8 \times 7 \times 5$ ways

f) 4 digit no. divisible by 5. find No of ways?

if Repetition allowed →

a	b	c	d	
↓	↓	↓	↓	0/5
⑨	⑩	⑩	②	No of cases = $9 \times 10 \times 10 \times 2$ ways

if Repetition not allowed →

a	b	c	d	
↓	↓	↓	↓	0
⑨	⑧	⑦	①	(fix zero) — No of cases = $9 \times 8 \times 7 \times 1$ ways

a	b	c	d	
↓	↓	↓	↓	5
⑨	⑧	⑦	①	(fix five) — + No of cases = $9 \times 8 \times 7 \times 1$ ways

Total = $9 \times 8 \times 7 \times 2$ ways

g) find No. of 5 digit no. whose sum of digits is even (Repetition allowed)

a	b	c	d	e	
9	10	10	10	10	→ No. of 5 digit no = 90,000 numbers
					= $9 \times 10 \times 10 \times 10 \times 10$

We know for n numbers, sum of digits = even → $\frac{n}{2}$ numbers
 sum of digits = odd → $\frac{n}{2}$ numbers

∴ No. of 5 digit no whose sum of digits is even = $\frac{90000}{2} = 45000$ numbers

* Divisibility of 4 : —

1) 0, 1, 2, 3, 4, 5, 6 → digits given
find 4 digit no. divisible by 4. (Repetition not allowed)

fix	5	4	0	4	→ 20 cases
Last	4	3	1	2	→ 12 cases
two	4	3	1	6	→ 12 cases
Digits	5	4	2	0	→ 20 cases
	4	3	2	4	→ 12 cases
	4	3	3	2	→ 12 cases
	4	3	3	6	→ 12 cases
	5	4	4	0	→ 20 cases
	4	3	5	2	→ 12 cases
+	4	3	5	6	→ 12 cases
+	5	4	6	0	→ 20 cases
+	4	3	6	4	→ 12 cases

Total no. of 4 digit no. with distinct digits divisible by 4 = 176

* Divisibility by 3 — (3 & 9 follow same rule)

Case 1 : given: 1, 2, 3, 4, 5 → find 4 digit no. divisible by 3
→ Remove one digit

$$1 + 2 + 3 + 4 + 5 = 15$$

$$15 - n \mod 3 = 0$$

∴ $n = 3$ ∴ Remaining → 1, 2, 4, 5 = $4! = 24$ numbers

Case 2 : 2, 3, 4, 5, 6 → find 3 digit no. divisible by 3
→ Remove 2 digits

$$2 + 3 + 4 + 5 + 6 = 20$$

$$20 - (x + y) \mod 3 = 0$$

- (2+6) → 3, 4, 5 → 3! ways = 6 Ans = 6+6+6 ways
 - (3+5) → 2, 4, 6 → 6 ways ∴ 18 numbers
 - (4+6) → 2, 3, 5 → 6 ways (Ans)

* Nothing's in Linear (we want to select r things s.t no two are consecutive)

Formula — $n - r + 1 C_r$

Ex: — 10 house Buglar choose → 4 house

<u>M1</u> $10 - 4 + 1 C_4$ $= 7 C_4$ $= \frac{7 \times 6 \times 5 \times 4}{1 \times 2 \times 3 \times 4}$ $= 35 \text{ ways}$	<u>M2</u> $\textcircled{n}$ ≥ 0 C_1 ≥ 1 $\textcircled{y}$ ≥ 1 C_2 ≥ 1 $\textcircled{z}$ ≥ 1 C_3 ≥ 1 $\textcircled{k}$ ≥ 1 C_4 ≥ 0 $\textcircled{l}$ ≥ 0 $x + y + z + k + l = 6$ $\Rightarrow x + y + z + k + l = 3$ $\rightarrow 3 + 5 - 1 C_{5-1} = 7 C_4$
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* Sum of all no. used at once from 1, 2, 3, 4.

Repetition not allow → $\frac{1n}{n}$ (sum of digit) (111... n times) $\left| \frac{1^4}{4} (1+2+3+4) (1111) \right|$

Repetition allow → $\frac{n^n}{n}$ (sum of digit) (111... n times) $\left| \frac{4^4}{4} (1+2+3+4) (1111) \right|$

Ex: — 0, 2, 4, 6, 8 → Sum of all 5 digit number?

$$\frac{1^5}{5} (0+2+4+6+8) (11111) - \frac{1^4}{4} (2+4+6+8) (1111)$$

(cases when 0 is in first place)

Follow this method when the digits include 0.

Ex: — 2, 3, 4, 5 → sum of all 4 digit numbers?

Repetition allowed — $\frac{4^4}{4} (2+3+4+5) (1111)$

Repetition not allow — $\frac{1^4}{4} (2+3+4+5) (1111)$

* Ring Concept

6 different rings $\longrightarrow$ 4 finger



1st Ring $\rightarrow$ 4 options
 2nd Ring $\rightarrow$ 5 options
 3rd Ring $\rightarrow$ 6 options
 4th Ring $\rightarrow$ 7 options
 5th Ring $\rightarrow$ 8 options
 6th Ring $\rightarrow$ 9 options
 Ans = $4 \times 5 \times 6 \times 7 \times 8 \times 9$

M2 ——— Consider similar $\rightarrow$ different formula

$$a + b + c + d = 6$$

$$\rightarrow 6 + 4 - 1 \text{ } {}^6C_4 - 1 = {}^9C_3$$

$$\text{Ans} = {}^9C_3 \times \underline{16}$$

(arrangement of rings)

$$\text{Ans} = \frac{9 \times 8 \times 7 \times \cancel{6}}{\underline{13} \times \cancel{6}} \times \underline{16}$$

$$\text{Ans} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times \cancel{3}}{\underline{13}}$$

* Derangement :

B_1, B_2, B_3, B_4, B_5 (5 Boys)

P_1, P_2, P_3, P_4, P_5 (5 Destination)

a) No one goes to his desized destination = $D(5)$

b) 2 reach destination, 3 did not reach Destination —

$$= {}^5C_3 \times D(3) \times {}^2C_2 \times 1$$

did not reach

2 person who reached

Now,

$$D(2) = 1$$

$$D(3) = 2$$

$$D(4) = 9$$

$$D(5) = 44$$

$$D(6) = 265$$

$$D(n) = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + \frac{1}{n!} \right)$$

Important formula :

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$$

Ex — Renu has 4 friends. She wants to invite atleast 4 friend in her party. find no. of ways

$${}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4$$

$$= {}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 - {}^4C_0 = 2^4 - 1 = 15 \text{ ways}$$

Concept $\rightarrow$ 5 Men, 4 Women

Select a Team of 4 people such that it contains at least 1 woman
 $\geq 1W$

M1 —

$$1W, 3M \rightarrow {}^4C_1 \times {}^5C_3 = 40$$

$$2W, 2M \rightarrow {}^4C_2 \times {}^5C_2 = 60$$

$$3W, 1M \rightarrow {}^4C_3 \times {}^5C_1 = 20$$

$$4W, 0M \rightarrow {}^4C_4 \times {}^5C_0 = 1$$

121 ways

M2 —

$$\text{atleast 1} = \text{Total} - \text{None}$$

$$= \text{Total} - \text{No women}$$

$$= {}^9C_4 - {}^5C_4$$

$$\begin{aligned} *) \text{Atleast 2} &= \text{Total} - (\text{none} + \text{Exactly 1}) \\ &= {}^9C_4 - ({}^5C_4 + {}^4C_1 \times {}^5C_3) \end{aligned}$$

*) Letter envelop Concept —

4 letter, 4 envelop. How many wrong choices can be made?

M1

1st letter — 4 option

2nd letter — 3 option

3rd letter — 2 option

4th letter $\rightarrow$ 1 option

$$\text{Total Possibility} = 4 \times 3 \times 2 \times 1 = 24$$

$$\text{Correct choice} = 1$$

$$\therefore \text{Wrong choice} = 23$$

M2

$$\text{Case 1) All four choice are wrong} \\ = D(4) = 9$$

$$\text{Case 2) 3 wrong \& 1 correct} \\ = {}^4C_3 \times D(3) \times 1C_1 \times 1 = 8$$

$$\text{Case 3) 2 wrong \& 2 correct} \\ = {}^4C_2 \times D(2) \times 2C_2 \times 1 = 6$$

$$\text{Case 4) 3 correct \& 1 wrong} \\ \rightarrow \text{invalid case (logical error)}$$

$$\therefore \text{Wrong cases} = 9 + 8 + 6 = 23$$

$$\begin{aligned} *) \text{How many integers in } \{100, 101, \dots, 999\} \text{ have } \underline{\text{at least 1 digit repeated}} \\ = \text{Total} - \text{no digit repeated} \\ = (9 \times 10 \times 10) - (9 \times 9 \times 8) \end{aligned}$$

*) VVIP :-

$$\boxed{x - y - z = 25} \text{ and } x \leq 40; y \leq 12; z \leq 12$$

$$\Rightarrow y + z = x - 25$$

$$\underline{x = 40}, y + z = 40 - 25 = 15$$

$$12 + 3$$

$$11 + 4$$

$$10 + 5$$

$$9 + 6$$

$$8 + 7$$

$$6 + 9$$

$$5 + 10$$

$$4 + 11$$

$$3 + 12$$

10 cases

$$\underline{x = 39} \rightarrow y + z = 39 - 25 = 14$$

$$12 + 2$$

$$11 + 3$$

$$10 + 4$$

$$9 + 5$$

$$8 + 6$$

$$7 + 7$$

$$6 + 8$$

$$5 + 9$$

$$4 + 10$$

$$3 + 11$$

$$2 + 12$$

11 cases

$$\underline{x = 38} \rightarrow y + z = 38 - 25 = 13$$

12 cases

$$\underline{x = 37}, y + z = 37 - 25 = 12$$

11 cases

$$\underline{x = 36}, y + z = 36 - 25 = 11$$

10 case

$$\underline{x = 35}, y + z = 35 - 25 = 10$$

9 case

$$\underline{x = 34} \rightarrow 8 \text{ Case}$$

$$\underline{x = 33} \rightarrow 7 \text{ case}$$

...

$$\underline{x = 27}, x + y = 2 \rightarrow 1 \text{ Case}$$

$$\therefore \text{Ans} = (1 + 2 + 3 + \dots + 12) + 10 + 11 \text{ cases}$$

* Numerical on Words / digits

$$A, B, C, D \rightarrow \text{ways of arranging} = 4! = \underline{4} = 24 \text{ ways}$$

$$A, A, C, D \rightarrow \text{ways of arranging} = \frac{4!}{2!} = \frac{4 \times 3 \times \cancel{2}}{\cancel{2}} = 12 \text{ ways}$$

ALLAHABAD $\rightarrow$ How many different words —

$$A \rightarrow 4$$

$$L \rightarrow 2$$

$$B \rightarrow 1$$

$$D \rightarrow 1$$

$$H \rightarrow 1$$

$$= \frac{4!}{4! 2! 1! 1! 1!}$$

Q) How many 4 Letter words can be formed using "ALLAHABAD"

$$\boxed{A \rightarrow 4}$$

$$\boxed{L \rightarrow 2}$$

$$H \rightarrow 1$$

$$B \rightarrow 1$$

$$D \rightarrow 1$$

1) All 4 letters different ($n y z k$)

$$= {}^5C_4 \times 4!$$

2) two letter same, two different ($n n y z$)

$$= {}^2C_1 \times {}^4C_2 \times \frac{4!}{2!}$$

3) three letter same, 1 different ($n n n y$)

$$= {}^1C_1 \times {}^4C_1 \times \frac{4!}{3!}$$

4) $\therefore$ Two same of one kind, Two same of different kind ($n n y y$)

$$= {}^2C_2 \times \frac{4!}{2! 2!}$$

5) All letters same — ($n n n n$)

$$= {}^1C_1 \times \frac{4!}{4!}$$

* "MATHEMATICS" → All vowels come together.
 Find no. of different ways —

$\boxed{A, E, A, I}$ M T H M T C S
 $\times \quad \checkmark \quad \times \quad \checkmark$

$$\text{Ans} = \frac{L^8}{L^2 L^2} \times \left(\frac{L^4}{L^2} \right)$$

(overall arrangement) (arrangement within block)

* How many 4 digit no are there, for which the product of digits is 10

$$a \quad b \quad c \quad d = 10$$

$$\begin{array}{cccc} 1 & \times & 1 & \times & 1 & \times & 10 \\ \hline 1 & \times & 1 & \times & 2 & \times & 5 \end{array} \longrightarrow \frac{L^4}{L^3} = 4 \text{ ways}$$

$$\longrightarrow \frac{L^4}{L^2} = 12 \text{ ways}$$

(> 5 per) (> 5 per) (> 5 per) (> 5 per) 16 ways

* How many 4 digit no are such that reading from right → left each digit is greater than digit at previous position.

$$a \quad b \quad c \quad d \text{ s.t. } a > b > c > d \mid 0, 1, 2, 3, 4, 5, 6, 7, 8, 9$$

$$\text{No. of ways} = {}^{10}C_4 \text{ (just pick 4 digit)}$$

$$a \quad b \quad c \quad d \text{ s.t. } a < b < c < d$$

$$\text{No. of ways} = {}^{10}C_4 - {}^9C_3$$

$$= {}^9C_4 \text{ (ans)}$$

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9$$

$$\text{Ex} - 1, 2, 0, 9$$

$$0, 1, 2, 9 \times \text{ (3 digit)}$$

not possible

do not pick 0

* Triangle Concept

of perimeter = 21. find no of triangle possible?

$$a+b+c=21; a \leq 10, b \leq 10; c \leq 10 \text{ (Max Value)} \\ \text{determine}$$

∴ Lets find ordered solution —

$$a = 10 - A \quad (10 - A) + (10 - B) + (10 - C) = 21$$

$$b = 10 - B$$

$$c = 10 - C$$

$$\Rightarrow \boxed{A+B+C=9} \rightarrow 9+3-1 \quad C_3-1 = {}^{11}C_2 = 55$$

$$\therefore \boxed{\text{ordered solution} = 55} \\ \Rightarrow 6x + 3y + z = 55$$

Now, find unordered solution —

a) 2 same, 1 different where $A=B$

$$2A + C = 9 \quad \therefore A = 0 \text{ to } 4$$

$$\Rightarrow C = 9 - 2A$$

$$\therefore \boxed{y = 5 - 1 = 4}$$

b) All same — $A=B=C$

$$A+A+A=21 \text{ (possible) when } A=7$$

$$\therefore \boxed{z = 1}$$

$$\text{Now, } 6x + 3y + z = 55$$

$$\Rightarrow 6x + (3 \times 4) + 1 = 55$$

$$\Rightarrow 6x = 55 - 13 = 42$$

$$\Rightarrow x = 7$$

total no of triangle

= unordered solution

$$= x + y + z = 7 + 4 + 1 = 12$$

No. of scalene triangle (x) = 7

No. of equilateral triangle (z) = 1

No. of isosceles triangle (y) = 4

Q) Another method —

If perimeter is odd

$$\text{No. of Triangle} = \frac{(P+3)^2}{48}$$

No. of scalene (Replace p by p-6)

$$\text{No. of scalene triangle} = \frac{(P-3)^2}{48}$$

Total Δ - scalene Δ = Iso Δ + equi Δ

Equilateral Triangle = 1 or 0
(Always)

If perimeter is even

$$\text{No. of Triangle} = \frac{P^2}{48}$$

No. of scalene (Replace P by P-6)

$$\text{No. of scalene triangle} = \frac{(P-6)^2}{48}$$

Total Δ - scalene Δ = Iso + equi

Equilateral Triangle = 1/0
(Always)

* Concept :

n different things divide into $\begin{matrix} P \\ \swarrow \\ r \text{ groups} \\ \searrow \\ r \end{matrix}$

$$\text{Then, No. of ways} = \frac{1^n}{1! 1! 1! \dots}$$

* P similar thing $\xrightarrow{\geq 0 \text{ select}}$ (P+1) ways

Ex: — there are 5 mangoes. find no. of ways of selecting 0 or more mangoes —
Ans = 6 ways.

Ex: — 5 similar Mangoes, 6 similar Apples & 4 different Banana.
find no. of way of picking atleast 1 Mango & 1 apple.

Ans — $5 \times 6 \times 2^4$

1M — 1A
2A
3A
4A
5A
6A
} 6 cases

1st M = 6 cases

2nd M = 6 cases

3rd M = 6 cases

4th M = 6 cases

5th M = 6 cases

Total = 5×6 cases

Banana — all possibility

$$({}^1C_0 + {}^1C_1 + {}^1C_2 + {}^1C_3 + {}^1C_4)$$

= 2^4 possibility

* Balls & Boxes Concept

Case 1) $\rightarrow$ Different Balls $\rightarrow$ Different Boxes

Case 2) $\rightarrow$ Similar Balls $\rightarrow$ Similar Boxes

Case 3) $\rightarrow$ Different Balls $\rightarrow$ similar Boxes

Case 4) $\rightarrow$ Different Balls $\rightarrow$ Different Boxes (restriction)

Case 5) $\rightarrow$ Similar $\rightarrow$ different

Case 1: — Different Balls $\rightarrow$ Different Boxes

Ex: — 3 balls & 5 Boxes

$$\text{No. of ways/possibilities} = 5 \times 5 \times 5 \\ = 125 \text{ possibility}$$

1st ball $\rightarrow$ 5 boxes
2nd ball $\rightarrow$ 5 boxes
3rd ball $\rightarrow$ 5 boxes

Case 4: — Different Balls $\rightarrow$ Different Boxes (restriction)

5 Different balls to 3 Different Box s.t no box is empty

$$\begin{array}{ccc} \underline{\quad} & \underline{\quad} & \underline{\quad} \end{array} \rightarrow \left(\frac{{}^5C_3 \times {}^2C_1 \times {}^1C_1}{12} \right) \times \begin{array}{l} 3 \text{ ways} \\ (\text{arrangement}) \end{array}$$

$$\begin{array}{ccc} \underline{\quad} & \underline{\quad} & \underline{\quad} \end{array} \rightarrow \left(\frac{{}^5C_2 \times {}^3C_2 \times {}^1C_1}{12} \right) \times \begin{array}{l} 3 \text{ ways} \\ (\text{arrangement}) \end{array}$$

Add these to obtain the total no. of ways.

Case 3: — Different Balls $\rightarrow$ Similar Boxes

5 Different balls $\rightarrow$ 3 Similar Boxes

$$1) \underline{5} \quad \underline{0} \quad \underline{0} \rightarrow {}^5C_5$$

$$2) \underline{4} \quad \underline{1} \quad \underline{0} \rightarrow {}^5C_4 \times {}^1C_1$$

$$3) \underline{3} \quad \underline{2} \quad \underline{0} \rightarrow {}^5C_3 \times {}^2C_2$$

$$4) \underline{3} \quad \underline{1} \quad \underline{1} \rightarrow ({}^5C_3 \times {}^2C_1 \times {}^1C_1 / 12)$$

$$5) \underline{2} \quad \underline{2} \quad \underline{1} \rightarrow ({}^5C_2 \times {}^3C_2 \times {}^1C_1 / 12)$$

Case 2) Similar Balls to Similar Boxes

6 similar Balls to 3 similar Boxes

$\Rightarrow$ just find unordered solution

$$a + b + c = 6 \rightarrow \text{ordered solution} = 6 + 3 - 1 C_{3-1} = {}^8C_2 = 28$$

Remove cases — This, $6x + 3y + z = 28$

a) 2 same & 1 different —

$$2a + c = 6 \quad a = 0 \text{ to } 3$$

$$\Rightarrow c = 6 - 2a \quad \therefore y = 9 - 1 = 3$$

b) All same —

$$a + a + a = 6 \text{ (possible)}$$

$$\therefore z = 1$$

$$\text{Thus, } 6x + (3 \times 3) + 1 = 28$$

$$\Rightarrow 6x = 28 - 10$$

$$\Rightarrow 6x = 18 \Rightarrow x = 3$$

$$\therefore \text{unordered sol} = x + y + z = 3 + 3 + 1 = 7 \text{ (Ans)}$$

Case 5) Similar to Different Balls Boxes

$\Rightarrow$ just find ordered solution

6 similar Balls to 3 different Boxes

$$a + b + c = 6$$

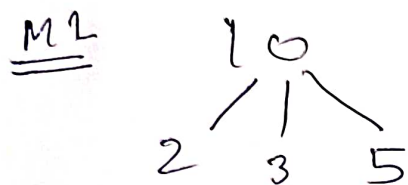
$$\rightarrow 6 + 3 - 1 C_{3-1} = {}^8C_2 = 28 \text{ (Ans)}$$

* Important variety —

Remember → Group/object → arrangement not required X

Person/children → arrangement required ✓

Ex: — 10 different objects $\begin{cases} 2 \text{ object} \\ 3 \text{ object} \\ 5 \text{ object} \end{cases}$ (3 groups)



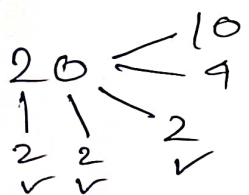
$$\text{Ans} = 10C_2 \times 8C_3 \times 5C_5$$

M2

$$n \begin{cases} p \\ q \\ r \end{cases} \quad \text{Ans} = \frac{1n}{1p \ 1q \ 1r}$$

$$\text{Ans} = \frac{110}{12 \ 13 \ 15}$$

* 20 objects divided in groups of 10, 4, 2, 2, 2 respectively.



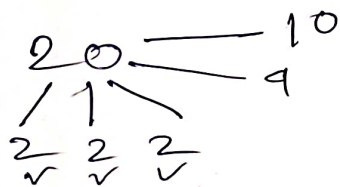
M1

$$\frac{20C_2 \times 10C_4 \times 6C_2 \times 4C_2 \times 2C_2}{13}$$

M2

$$\frac{1}{13} \left(\frac{20}{110 \ 14 \ 12 \ 12 \ 12} \right)$$

* 20 children divided in groups of 10, 4, 2, 2, 2



M1

$$\frac{20C_2 \times 10C_4 \times 6C_2 \times 4C_2 \times 2C_2}{13} \times \frac{1}{5} \quad (\text{Arrangement of groups})$$

M2

$$\left(\frac{120}{12 \ 12 \ 12 \ 14 \ 110} \right) \times \frac{1}{13} \times \frac{1}{5} \quad (\text{Arrangement of groups})$$

* Probability : —————

1) $\text{Probability} = \frac{\text{favourable outcomes/cases}}{\text{Total outcomes/cases}}$

$$P = \frac{n(A)}{n(S)} ; \boxed{0 \leq P \leq 1}$$

2) Sample space = The set of all elements

$$1 \text{ Die} = \{1, 2, 3, 4, 5, 6\}$$

$$1 \text{ Coin} = \{H, T\}$$

$$2 \text{ Coin} = \{HH, TT, HT, TH\}$$

3) $\frac{\text{favourable event}}{\text{unfavourable event}} = 1$

$$4) P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

5) Mutually Exclusive Event : ——— $A \cap B = \phi$
(one event impact the other)

6) Independent Event : ——— $P(A \cap B) = P(A) \times P(B)$
(one event doesn't impact the other event)

7) Mutually Exhaustive event : — $A \cup B = \text{Sample space}$
where A & B are two events.

8) Picking = Selection only (Combination ${}^n C_r$)

9) favourable - unfavourable Expectancy / Probability : —

$$P = {}^n C_r \times p^r \times q^{n-r}$$

$n = \text{Total cases}$	$p^r = \text{Probability of favourable}$	$q^{n-r} = \text{Probability of unfavourable}$
$r = \text{selected/favourable cases}$		

*) Total Probability Concept : —

<u>B1</u>	<u>B2</u>
5 W	6 W
6 R	5 R

find Probability to get a white Ball

Ans — Case 1 : — Bag 1 — $\frac{1}{2} \times \frac{5}{11}$ (Total Balls = 11, white = 5)
(Prob of Bag selection)

Case 2 : — Bag 2 — $\frac{1}{2} \times \frac{6}{11}$ (Total Balls = 11, white = 6)

$$\therefore \text{Total Probability} = \left(\frac{1}{2} \times \frac{5}{11} \right) + \left(\frac{1}{2} \times \frac{6}{11} \right) = \frac{1}{2} \text{ (Ans)}$$

*) Bayes Theorem : —

<u>B1</u>	<u>B2</u>	<u>B3</u>
2 W	1 W	3 W
3 R	4 R	5 R

A Red Ball is selected. find probability that selected ball is from Bag 2.

$$\text{Probability} = \frac{\text{Favourable Probability}}{\text{Total Probability}} = \frac{\frac{1}{3} \times \frac{4}{5}}{\left(\frac{1}{3} \times \frac{3}{5} \right) + \left(\frac{1}{3} \times \frac{4}{5} \right) + \left(\frac{1}{3} \times \frac{5}{8} \right)}$$

$$*) P(A|B) = \frac{P(A) \cdot P(B|A)}{P(B)} \quad \left| \begin{array}{l} P(A|B) = \text{prob of } A \text{ given } B \\ P(B|A) = \text{prob of } B \text{ given } A \end{array} \right.$$

*) important variety : — (Cricket/football match)

$$P(C) = 1/6, \quad P(\text{atleast one match win}) \geq \frac{1}{2} \text{ (Given)}$$

$$P(\bar{C}) = 5/6$$

$$\text{Ans: — } P(\text{atleast 1 match win}) \geq \frac{1}{2}$$

$$\Rightarrow \text{Total — } P(\text{No Match win}) \geq \frac{1}{2}$$

$$\therefore P(\text{Wo. match win}) \leq \frac{1}{2}$$

check with 5/6

then check $5/6 \times 5/6$

then check $5/6 \times 5/6 \times 5/6$ & so on